

EDUCATION	Massachusetts Institute of Technology <i>Ph.D. in Electrical Engineering and Computer Science</i> Cambridge, MA Sep 2025 – ??
	University of Massachusetts Amherst <i>B.S. in Computer Science, summa cum laude</i> Sep 2021 – May 2025 GPA: 4.0, Member of Commonwealth Honors College Minors: Philosophy, Psychology Amherst, MA
RESEARCH INTERESTS	Evolutionary & Reinforcement Learning · Open Endedness · Program Synthesis · Robotics · Cognitive Models
RESEARCH EXPERIENCE	<i>Flowers Team, INRIA Bordeaux</i> 3/2024– Institut National de Recherche en Informatique et en Automatique Bordeaux, France Advisors: Clément Romic, Cédric Colas, Pierre-Yves Oudeyer → Developing novel mechanisms to dynamically balance exploration and exploitation in autotelic reinforcement learning systems. <i>Programs Under Selection and Heredity (PUSH) Lab</i> 9/2021– University of Massachusetts Amherst & Amherst College Amherst, MA Advisor: Lee Spector → Applying lexibase selection to evolutionary computation systems such as genetic programming and evolutionary reinforcement learning, with a focus on exploration and diversity. <i>Safe, Confident and Aligned Learning + Robotics (SCALAR) lab</i> 9/2022– University of Massachusetts Amherst Amherst, MA Advisor: Scott Niekum → Improving safety and alignment of reinforcement learning from human feedback (RLHF) systems through learning a diverse distribution of reward functions or policies from preferences with hidden context. <i>Advanced Agent – Robotics Technology (AART) Lab</i> 5/2024 – 12/2024 Carnegie Mellon University Pittsburgh, PA Advisor: Katia Sycara <i>Interactive and Collaborative Autonomous Robotics (ICAROS) lab</i> 5/2023-3/2024 University of Southern California Los Angeles, CA Advisor: Stefanos Nikolaidis
PUBLICATIONS	Journal Articles under “Ryan Boldi” and “Ryan Bahlous-Boldi” Ryan Boldi*, Martin Briesch*, Dominik Sobania, Alexander Lalejini, Thomas Hel-muth, Franz Rothlauf, Charles Ofria, and Lee Spector. 2023. Informed Down-Sampled Lexibase Selection: Identifying productive training cases for efficient problem solving. https://arxiv.org/abs/2301.01488. In <i>Evolutionary Computation</i>. MIT Press. * denotes equal contribution Book Chapters Lee Spector, Ryan Bahlous-Boldi, Paul Vander Vort, and Niko Lorantos. 2025. Evolution of Artificial Intelligence, Continued. In <i>Genetic Programming Theory and Practice XXII</i>. New York: Springer. To appear.

Lee Spector, Li Ding and **Ryan Boldi**. 2024. Particularity. In Genetic Programming Theory and Practice XX. New York: Springer. https://doi.org/10.1007/978-981-99-8413-8_9. Preprint posted at <https://arxiv.org/abs/2306.06812>

Conference and Workshop Papers

Ryan Bahlous-Boldi, Li Ding, Lee Spector, and Scott Niekum. 2025. Pareto-Optimal Learning from Preferences with Hidden Context. In the Reinforcement Learning Conference 2025. Posted at <https://arxiv.org/abs/2406.15599>.

Ryan Bahlous-Boldi*, Maxence Faldor*, Luca Grillotti, Hannah Janmohamed, Lisa Coiffard, Lee Spector, and Antoine Cully. 2025. Dominated Novelty Search: Rethinking Local Competition in Quality-Diversity. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '25). Posted at <https://arxiv.org/abs/2502.00593>

Ryan Boldi, Ashley Bao, Martin Briesch, Thomas Helmuth, Dominik Sobania, Lee Spector, Alexander Lalejini. 2024. Untangling the Effects of Down-Sampling and Selection in Genetic Programming. In Proceedings of ALIFE 2024: the 2024 Conference on Artificial Life. (ALIFE '24). https://doi.org/10.1162/isal_a_00832. MIT Press. Posted at <https://arxiv.org/abs/2304.07089>.

Ryan Boldi, Li Ding and Lee Spector. 2023. Objectives Are All You Need: Solving Deceptive Problems Without Explicit Diversity Maintenance. In the Workshop on Agent Learning in Open-Endedness at NeurIPS (ALOE @ NeurIPS 2023). Posted at <https://arxiv.org/pdf/2311.02283>

Ryan Boldi and Lee Spector. 2023. Can the Problem-Solving Benefits of Quality Diversity Be Obtained Without Explicit Diversity Maintenance? In Genetic and Evolutionary Computation Conference Companion (GECCO '23).

Ryan Boldi, Thomas Helmuth, and Lee Spector. 2022. The environmental discontinuity hypothesis for down-sampled lexicase selection. In The 2022 Conference on Artificial Life - Why it Didn't Work-Shop (ALIFE '22). Posted at <https://arxiv.org/pdf/2205.15931>

Li Ding, **Ryan Boldi**, Thomas Helmuth, and Lee Spector. 2022. Lexicase selection at scale. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '22).

Posters and Poster Papers

Ryan Boldi, Li Ding and Lee Spector. 2023. Solving Deceptive Problems without Explicit Diversity Maintenance. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24).

Ryan Boldi, Matthew Fontaine, Sumeet Batra, Gaurav Sukhatme and Stefanos Nikolaidis. 2024. Generating Diverse Induced Policies for Conditioned Policy Distillation. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24)

Ryan Boldi, Charles Zhang, Lee Spector. 2023. Encouraging Diversity in Reinforcement Learning with Lexicase Selection. RL at Harvard Workshop 2023.

Ryan Boldi, Ashley Bao, Martin Briesch, Thomas Helmuth, Dominik Sobania, Lee

Spector, Alexander Lalejini. 2023. The Problem Solving Benefits of Down-Sampling Vary by Selection Scheme. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO ‘23).

Ryan Boldi, Alexander Lalejini, Thomas Helmuth, Lee Spector. 2023. A static analysis of informed down-samples. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO ‘23).

Li Ding, **Ryan Boldi**, Thomas Helmuth, and Lee Spector. 2022. Going faster and hence further with lexicase selection. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO ‘22).

PRE-PRINTS

Ryan Boldi*, Aadam Lokhandwala*, Edward Annatone, Yuval Schechter, Alexander Lavrenenko, Cooper Sigrist. 2023. Improving Recommendation System Serendipity Through Lexicase Selection. Posted at <https://arxiv.org/abs/2305.11044>

TEACHING AND MENTORSHIP

Teacher, *X-Camp Academy* 9/2021–Present
 ➔ Mentored over 500 students (approx. 125 annually since 9/2021) in intensive USA Computing Olympiad (USACO) preparation courses, with many students advancing to higher divisions and achieving national-level recognition.
 ➔ Spearheaded the design, development, and instruction of X-Camp’s inaugural Artificial Intelligence (AI) curriculum, scaling the program from an initial 5 students to over 30 students per cycle and culminating in diverse student-led AI projects.

Teaching Research Team, *X-Camp Academy* 6/2022–Present
 ➔ Led the strategic migration of X-Camp’s academic operations (15+ instructors, 15+ courses, 500+ students) to a new ed-tech platform, directly enabling organizational growth to ≈30 instructors and nearly doubling student enrollment.
 ➔ Manage teacher recruitment, interviewing, and onboarding; define instructional guidelines to maintain high teaching quality and support the academy’s significant expansion.

President, *MassAI - UMass Amherst’s AI Student Organization* 1/2023–1/2025
 Team Leader, *Team UMass: ProjectX ML Research Comp. Winners* 9/2022–1/2023

PRESENTATION

(*In addition to those listed as conference/workshop papers and posters above*)

Conference

Encouraging Diversity in Reinforcement Learning with Lexicase Selection
 Poster: RL at Harvard Workshop 2023 Cambridge, MA

Think Before You Act: Generating High-Quality Diverse Reasoning Policies
 Poster: SoCal Undergraduate Research Symposium 2023 Los Angeles, CA

The Emergence of Diversity
 Emerging Researchers in Artificial Life Lightning Talk
 2023 Conference on Artificial Life Sapporo, Japan

Invited

Evolutionary Computation Spring 2023
 UMass Amherst Guest Lecture Amherst, MA
 COMPSI 389 - Introduction to Machine Learning

	Lexicase Selection and Reinforcement Learning Personal Autonomous Robotics Lab (PeARL), UT Austin Autonomous Learning Laboratory, UMass Amherst	Fall 2022 Austin, Texas Amherst, MA
	Lexicase Selection and the Diversity of Quality Adaptive and Intelligent Robotics Lab, Imperial College London	Summer 2022 London, UK
	Evolutionary Algorithms United Arab Emirates Ministry of Artificial Intelligence	Fall 2020 Dubai, UAE
AWARDS	<i>NSF Graduate Research Fellowship (GRFP)</i> National Science Foundation, 2025–2030	\$159,000
	<i>21st Century Leaders Award</i> University of Massachusetts Amherst, 2025	
	<i>Outstanding Undergraduate Achievement Award</i> Manning College of Information and Computer Sciences, 2025	
	Selected as an <i>HRI Pioneer</i> HRI'25: IEEE/ACM International Conference on Human-Robot Interaction	
	<i>Goldwater Scholarship</i> Barry Goldwater Scholarship & Excellence in Education Foundation, 2024	\$7,500
	<i>ProjectX ML Research Competition Winner</i> University of Toronto, 2023	\$20,000
	University of Massachusetts Amherst <i>21st Century Leaders Award</i> (2025) <i>Outstanding Undergraduate Achievement Award</i> (2025) <i>Alumni Achievement Scholarship</i> (2024) <i>Dean's Merit Scholarship</i> (2022) <i>John E. and Alice M. Flynn Scholarship</i> (2022)	\$750 \$1,500 \$3,300
MEMBERSHIP	<i>International Society for Artificial Life</i> <i>ACM SIGEVO, Special Interest Group for Genetic and Evolutionary Computation</i> <i>IEEE Robotics & Automation Society</i>	
SERVICE	<i>Reviewer, ALIFE 2025</i> <i>Reviewer, ICLR 2025</i> <i>Volunteer, RLC 2024</i> <i>Reviewer, GECCO 2024 Graph-based GP Workshop</i> <i>Volunteer, GECCO 2023</i>	<i>Kyoto, Japan</i> <i>Singapore</i> <i>Amherst, MA</i> <i>Melbourne, Australia</i> <i>Lisbon, Portugal</i>
COMPUTER SKILLS	<i>Languages & Frameworks</i> Python, Clojure, C++, Java, JavaScript, R, Numpy, PyTorch, Jax, Flax	